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Data Engineering – Hackathon 2

The second hackathon presents a curated set of challenges spanning SQL Queries, Business Insights, and Data Visualization. Below is the problem distribution across difficulty levels.

Category	Simple	Medium	Complex	Total Problems
SQL Queries	—	—	3	3
Business Insights	—	—	—	2
Data Visualization	—	4	—	4
Total Problems	—	4	3	7

 Table 2: Second Set of Problems (Optional) — These challenges are designed to test advanced SQL proficiency, analytical thinking, and data visualization skills.

Hackathon 2 — SQL Data Queries

Complex SQL

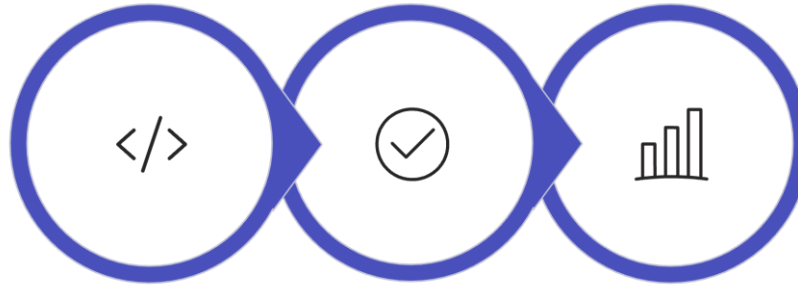
3 advanced query challenges requiring multi-table JOINS, aggregations, and ordering logic.

Data Visualization

4 visualization problems requiring structured SQL output ready for Excel charts and pivot tables.

Business Insights

2 insight-driven problems focused on customer behavior and revenue analysis.



Write SQL

**Validate
Output**

**Visualize
Results**

Each problem follows a structured approach — from writing the query to validating results against expected output and finally visualizing the data for business decision-making.

COMPLEX PROBLEM C1

Find the Peak Revenue Time Slots for Ticket Sales

Problem Statement: Write a query to determine which time slots generate the highest revenue. The output should include `show_datetime`, `total_revenue`, and `total_tickets_sold`.

Expected Output

	show_datetime	total_revenue	total_tickets_sold
►	2025-02-04 21:55:00	90818.73	147
	2025-03-09 15:08:00	81573.93	135
	2025-01-15 12:40:00	80438.77	139
	2025-03-21 06:57:00	71840.30	137
	2025-03-26 18:52:00	70527.02	133

COMPLEX PROBLEM C2

Identify Customers Who Purchased Both Tickets and Food

Problem Statement: Write a query to fetch customers who have purchased both movie tickets and food orders. The output should include user_id, name, total_spent_on_tickets, and total_spent_on_food.

Expected Output

	user_id	name	total_spent_on_tickets	total_spent_on_food
▶	4413	Jessica Wright	12931.77	583.75
	4727	Kathleen Reyes	12729.30	431.90
	366	Jimmy Stephens	12064.47	495.15
	4652	Margaret Jimenez	12170.28	313.20
	1567	Kathryn Stephens	12101.94	271.10
	3047	Brittany Green	11696.76	638.95
	1582	Katie Bass	11553.30	467.50
	2150	Tammy Valencia	11298.54	520.10
	1416	David Mcdonald	11201.76	539.75
	2925	Joshua Payne	11202.48	504.25

COMPLEX PROBLEM C3

Find the Most Profitable Screens Based on Ticket Sales

Problem Statement: Write a query to identify which screens have generated the most revenue from ticket sales. The output should include `screen_id`, `screen_name`, `total_tickets_sold`, and `total_revenue`.

Expected Output

	screen_id	screen_name	total_tickets_sold	total_revenue
▶	2	B	836	459610.99
	3	C	593	319182.97
	1	A	243	130773.12
	4	D	177	101719.11



Hackathon 2 — Data Visualization Challenges

The Data Visualization section challenges participants to write SQL queries whose output is structured for direct use in Excel — enabling bar charts, pie charts, and line charts that surface actionable business insights from cinema operations data.

VISUALIZATION PROBLEM M1

Screen Performance by Booking Volume

Problem Statement

Retrieve the total number of bookings for each screen. The data should be structured to allow visualization in Excel as a bar chart comparing the performance of different screens based on booking volume.

Business Insight

- Identify the most and least utilized screens
- Inform decisions about screen allocation for different movies and showtimes
- Potentially identify screens that may need upgrades or renovations based on usage

Visualization Suggestion

Bar Chart: Screen Name on the X-axis, Total Bookings on the Y-axis.

Pivot Table: Summarize the total bookings per screen name.

VISUALIZATION PROBLEM M2

Booking Delivery Method Distribution

Problem Statement

Retrieve the count of bookings for each delivery method specified in the tickets. The data should be structured to allow visualization in Excel as a pie chart showing the proportion of each delivery method used by customers.

Business Insight

- Understand customer preferences for ticket delivery methods
- Optimize delivery infrastructure and support based on popular choices
- Identify opportunities to encourage the adoption of more cost-effective delivery methods

Visualization Suggestion

Pie Chart: Each slice represents a delivery method, with the size proportional to the number of bookings. Display percentages for each slice.

Pivot Table: Summarize the count of bookings for each delivery method.

VISUALIZATION PROBLEM M3

Movie Revenue Analysis by Genre

Problem Statement

Retrieve the total revenue generated for each movie genre. The data should be structured to allow visualization in Excel as a bar chart showing revenue per genre.

Business Insight

- Identify which genres generate the most revenue
- Help plan future movie selections based on financial performance
- Support strategic decisions on genre-based marketing campaigns

Visualization Suggestion

Bar Chart: Movie genres on X-axis, Total Revenue on Y-axis.

Pivot Table: Summarize total revenue per genre.

VISUALIZATION PROBLEM M4

Monthly Ticket Sales Trend

Problem Statement

Retrieve monthly ticket sales data for the past 12 months, showing the total number of tickets sold each month. The output should be structured to create a line chart in Excel, showing the trend of ticket sales over time.

Business Insight

- Identify peak and low-demand months for strategic marketing and pricing
- Understand seasonal trends in ticket sales
- Support capacity planning and promotional scheduling

Visualization Suggestion

Line Chart: Month on X-axis, Tickets Sold on Y-axis.

Pivot Table: Summarize ticket sales per month.

Hackathon 2 — Key Takeaways



Advanced SQL Mastery

Multi-table JOINS, aggregations with GROUP BY, COALESCE for NULL handling, and LIMIT for top-N queries are core skills demonstrated across all three complex problems.



Data Visualization Readiness

Each visualization query is designed to produce output that maps directly to Excel charts — bar, pie, and line — enabling immediate business storytelling without additional transformation.



Business Insight Orientation

Every problem is grounded in real business questions — from identifying peak revenue slots to understanding delivery preferences — bridging the gap between raw data and strategic decisions.

THANK YOU

We value your time. Thank you for participating in the SRM
Talent Cloud Hackathon 2.

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